

# 3-Adapter Merging Report

## 1) Training of Models

Three LoRA adapters were trained independently, each on a different dataset, using the same base model.

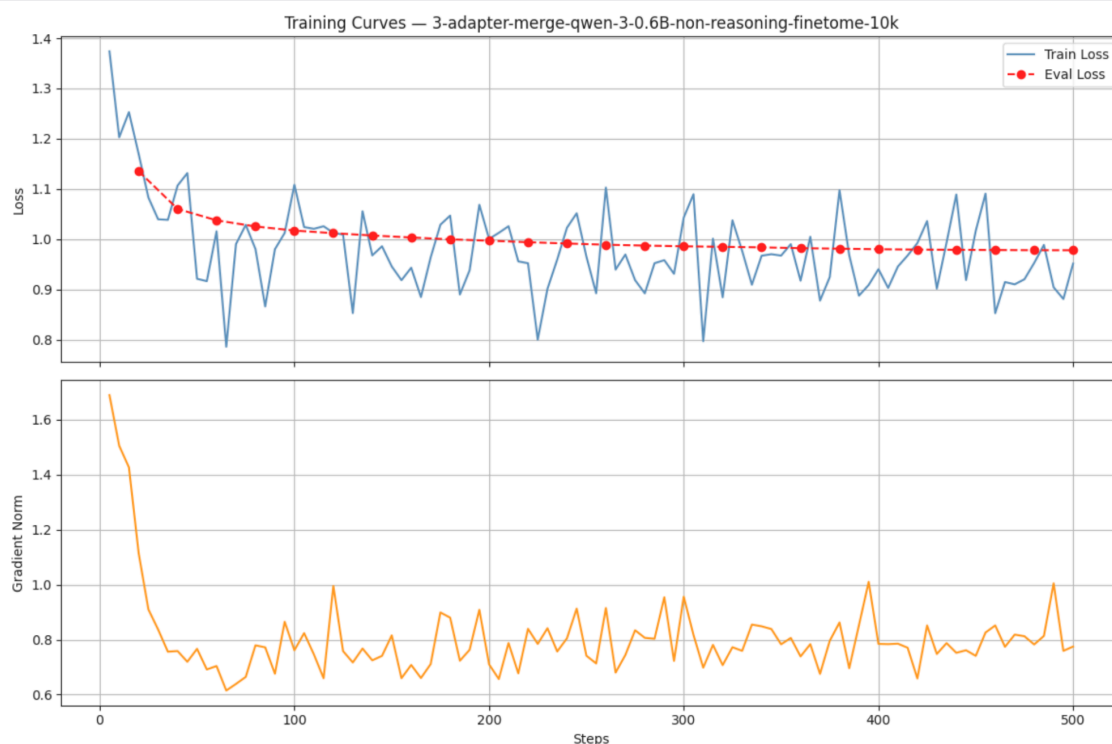
### Datasets Used:

- AG News — a news topic classification dataset covering four categories: World, Sports, Business, and Science/Technology. This dataset trains the model to recognize and label short news passages.
- OpenMath Reasoning — a mathematical reasoning dataset used to teach the model step-by-step problem solving and structured numerical reasoning.
- FineTome — a general-purpose instruction-following and conversational dataset, used to train broad, non-reasoning chat-style behaviour.

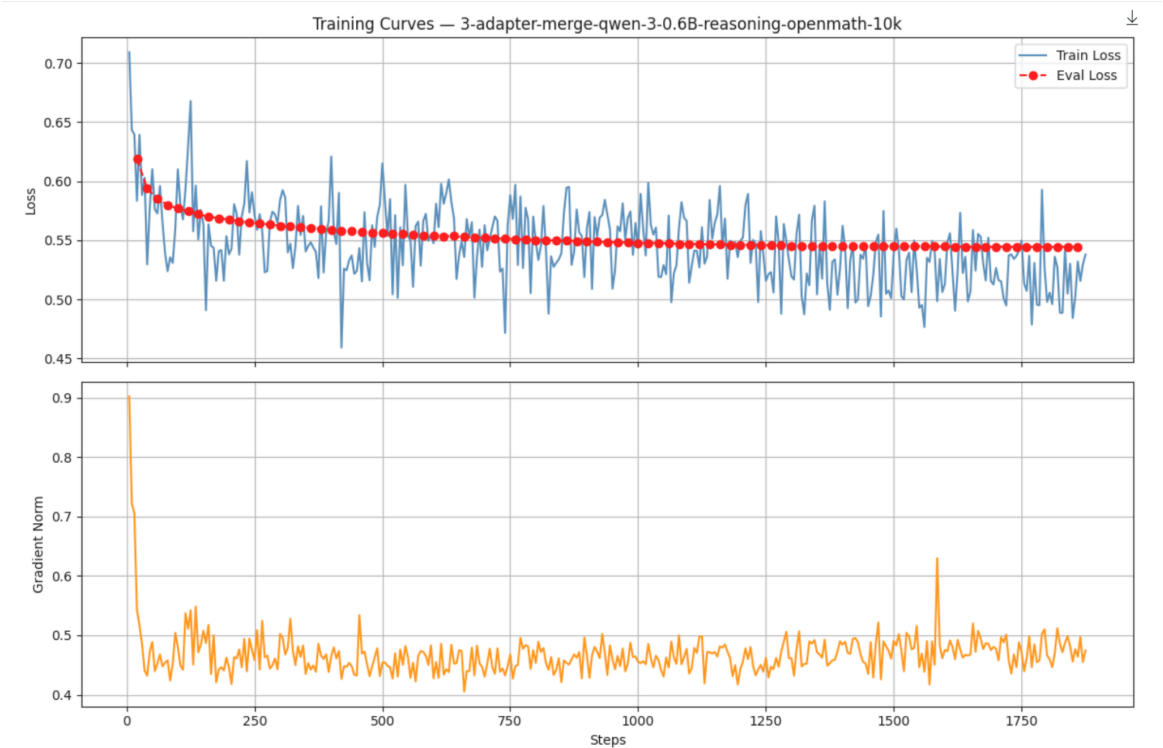
### Model Used

Qwen 3 0.6B — a 0.6-billion-parameter causal language model, used as the frozen base model onto which each LoRA adapter was trained.

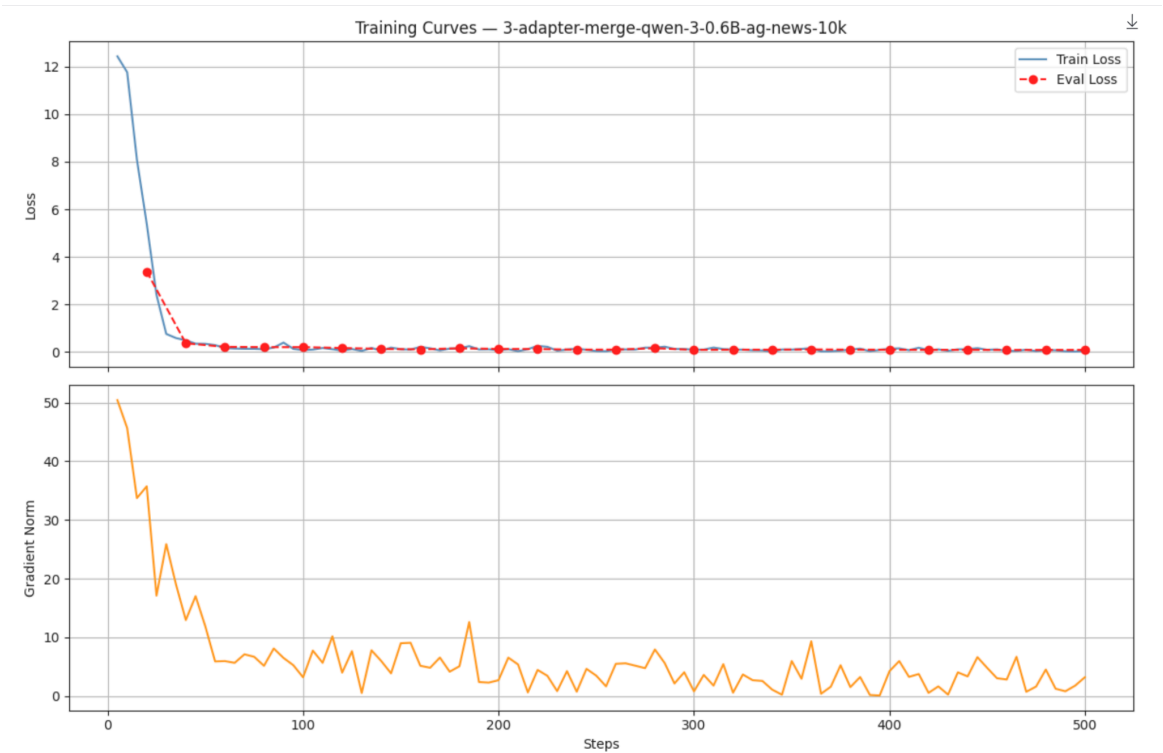
### Loss Curve for FineTome



# Loss Curve for OpenMath

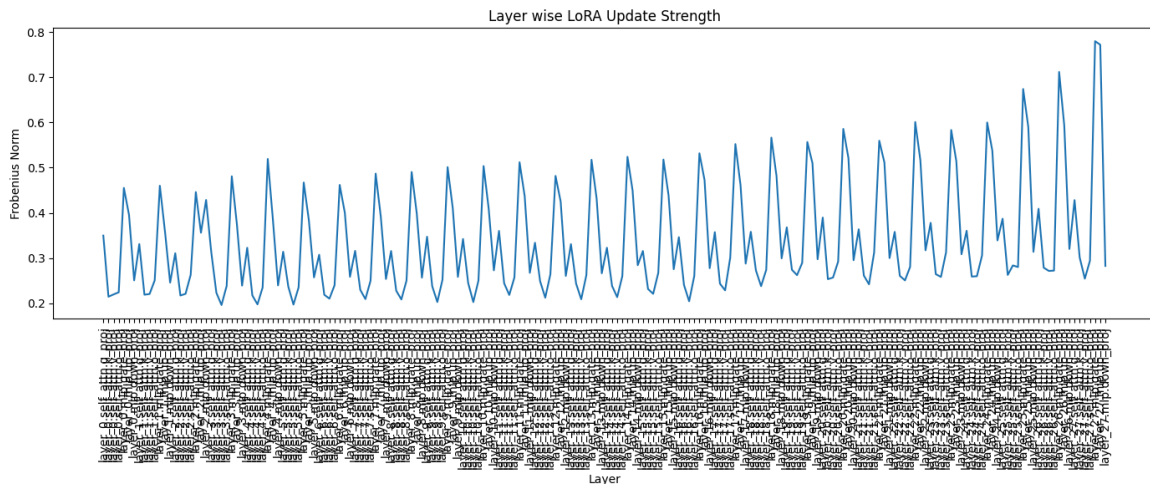


# Loss Curve for AG News



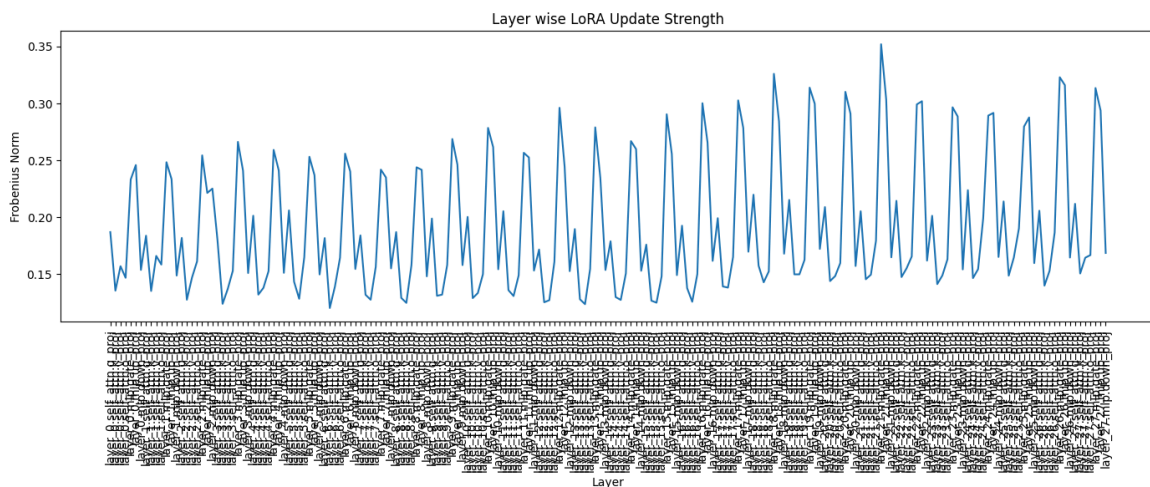
## 2) Comparison of Models

### Frobenius Norm of OpenMath Model



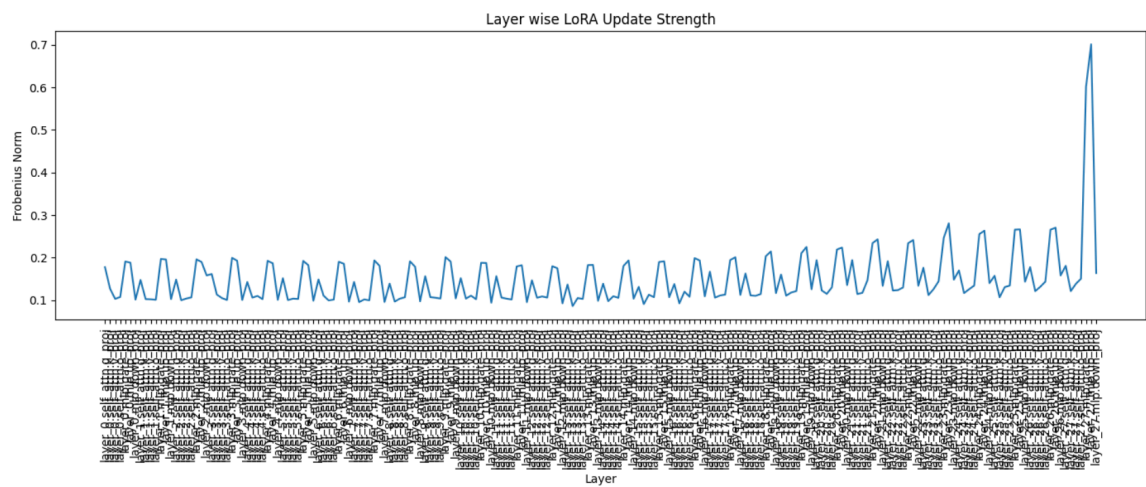
The OpenMath adapter's update strength ranges from about 0.2 to 0.8 and trends gradually upward across the depth of the network, with the largest peaks (around 0.6–0.78) occurring in the later layers. This suggests that learning mathematical reasoning required comparatively larger, more concentrated adjustments in the deeper layers of the model.

### Frobenius Norm of FineTome Model



The FineTome adapter shows a much smaller overall range, roughly 0.13 to 0.35, with a similar mild upward trend across layers. The consistently lower magnitude compared to OpenMath suggests that general instruction-following required lighter, more distributed adjustments rather than large, concentrated changes.

## Frobenius Norm of AG News Model



Most layers in the AG News adapter show small, fairly flat updates in the 0.1–0.3 range, except for one pronounced spike near the very end of the network, reaching about 0.7. This pattern is consistent with a classification task: most of the representational work is left untouched, while a small number of late layers closer to the output absorb a disproportionately large share of the adjustment needed to produce class labels.

## Cosine Similarity Between Adapters

### *OpenMath vs. FineTome (sample layers)*

Cosine similarity between the OpenMath adapter and the FineTome adapter was positive across every sampled layer and projection, though generally modest in size:

| Layer                     | Cosine Similarity |
|---------------------------|-------------------|
| layer_0.mlp.down_proj     | 0.0533            |
| layer_0.mlp.gate_proj     | 0.0408            |
| layer_0.mlp.up_proj       | 0.0780            |
| layer_0.self_attn.k_proj  | 0.0696            |
| layer_0.self_attn.o_proj  | 0.0734            |
| layer_0.self_attn.q_proj  | 0.0762            |
| layer_0.self_attn.v_proj  | 0.1441            |
| layer_1.self_attn.v_proj  | 0.2251            |
| layer_10.self_attn.v_proj | 0.2549            |
| layer_10.self_attn.q_proj | 0.1421            |
| layer_10.self_attn.k_proj | 0.1323            |

### ***AG News vs. Other Adapters (sample layers)***

By contrast, cosine similarity scores between the AG News adapter and the other adapters were close to zero and frequently negative:

| Layer                    | Cosine Similarity |
|--------------------------|-------------------|
| layer_0.mlp.down_proj    | 0.0019            |
| layer_0.mlp.gate_proj    | 0.0085            |
| layer_0.mlp.up_proj      | -0.0018           |
| layer_0.self_attn.k_proj | 0.0026            |
| layer_0.self_attn.o_proj | 0.0015            |
| layer_0.self_attn.q_proj | 0.0220            |
| layer_0.self_attn.v_proj | -0.0038           |
| layer_1.self_attn.v_proj | 0.0350            |
| layer_10.mlp.down_proj   | -0.0074           |
| layer_10.mlp.gate_proj   | -0.0014           |
| layer_10.mlp.up_proj     | 0.0007            |

Values for AG News sit an order of magnitude lower than the OpenMath/FineTome comparison and switch sign between layers. This indicates the AG News adapter learned an update direction that is largely independent (close to orthogonal) from the reasoning and instruction-following adapters.

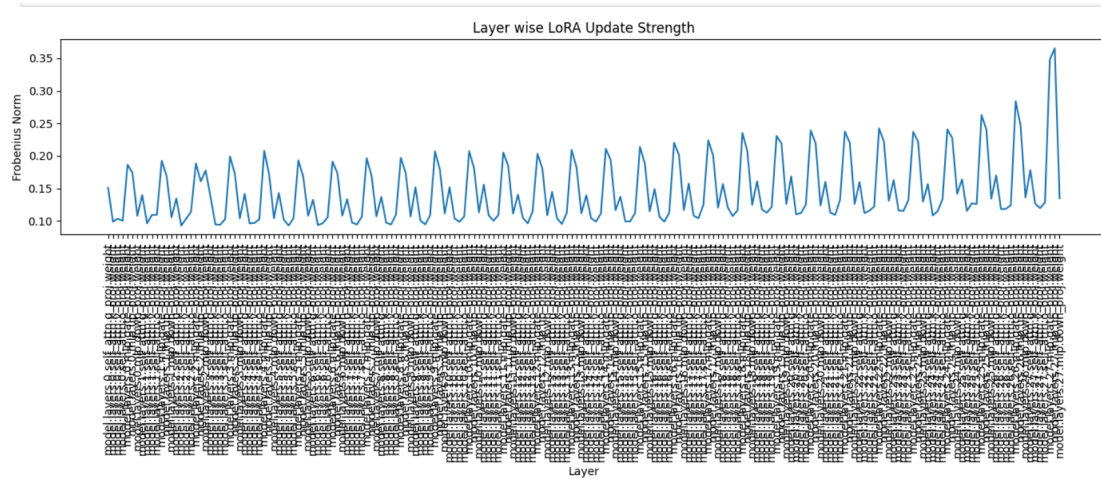
## **3) Merging of Models**

The three trained adapters (AG News, OpenMath, FineTome) were combined into a single merged adapter using five different merging methods. Each method represents a different strategy for resolving overlaps and conflicts between the individually trained updates.

### **Linear**

Linear merging simply averages the weight updates from all three adapters, giving each one an equal (or weighted) contribution to the final merged update.

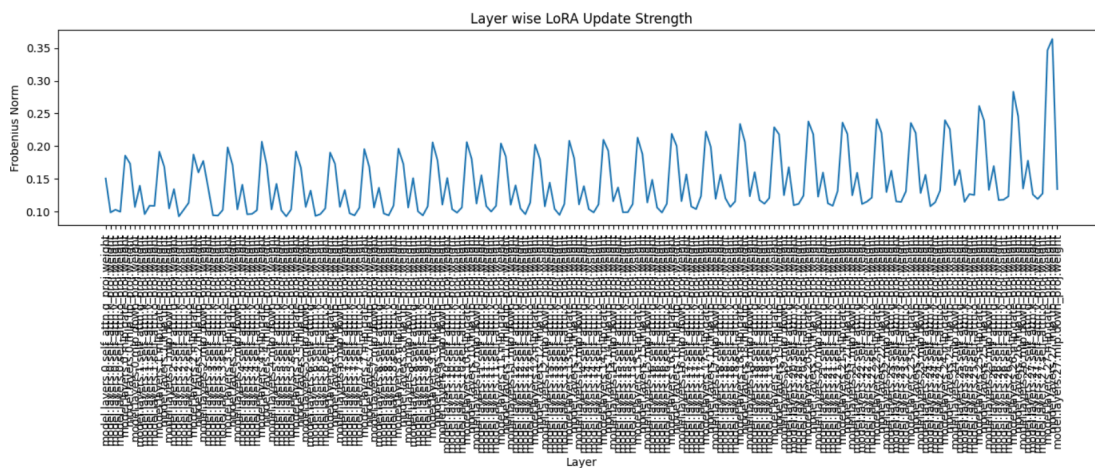
Frobenius norm:



## SVD

SVD-based merging performs a singular value decomposition across the stacked adapter updates to identify their principal shared directions, then reconstructs the merged adapter from those compressed components.

Frobenius norm:



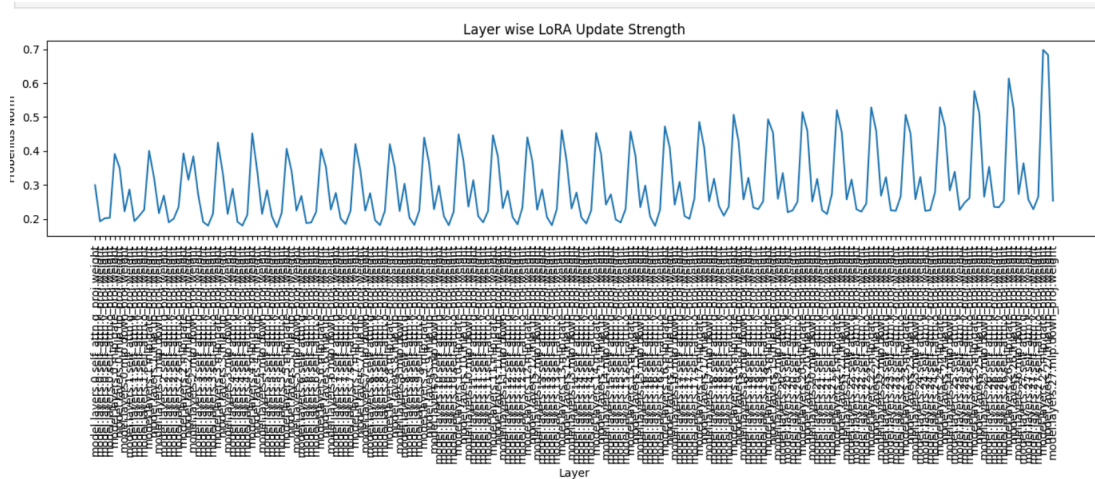
*Layer-wise LoRA Update Strength — SVD merge*



## TIES

TIES merging keeps only the most significant parameter changes from each adapter, resolves sign conflicts where adapters disagree on the direction of a change, and then combines the surviving, agreed-upon updates.

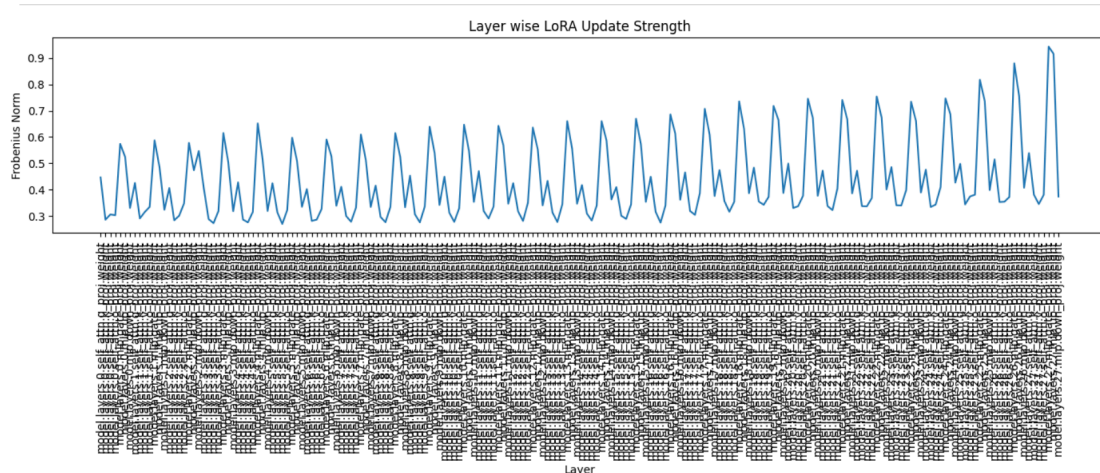
Frobenius norm:



## DARE

DARE randomly drops a fraction of the update values in each adapter and rescales the remaining values before combining them, which is intended to reduce interference between adapters while preserving their overall effect.

Frobenius norm:

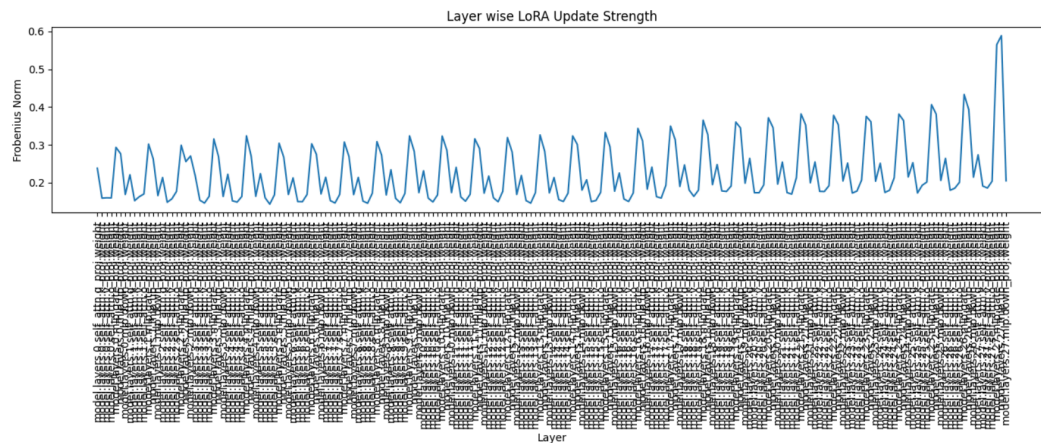


*Layer-wise LoRA Update Strength — DARE merge*

## SLERP

SLERP (spherical linear interpolation) blends adapters by interpolating smoothly along the direction between them on a sphere, rather than simply averaging their magnitudes.

Frobenius norm:



## 4) Evaluation Results

### Evaluated on OpenMath and FineTome

Each merged model was evaluated on the reasoning/instruction tasks using Exact Match(for openmath) and ROUGE-1 / ROUGE-2 / ROUGE-L scores(finetome):

| Model                    | Exact  | R1     | R2     | RL     |
|--------------------------|--------|--------|--------|--------|
| linear-3-adapters-merged | 0.1200 | 0.3088 | 0.0623 | 0.1465 |
| svd-3-adapters-merged    | 0.1400 | 0.3090 | 0.0626 | 0.1488 |
| ties-3-adapters-merged   | 0.0900 | 0.3294 | 0.0739 | 0.1595 |
| dare-3-adapters-merged   | 0.1050 | 0.3264 | 0.0721 | 0.1578 |
| slerp-3-adapters-merged  | 0.0900 | 0.3160 | 0.0676 | 0.1529 |
| Finetome adapter         | 0.0750 | 0.3360 | 0.0816 | 0.1671 |
| Openmath adapter         | 0.1350 | 0.3127 | 0.0688 | 0.1517 |



## Evaluated on AG News

Each merged model was also evaluated as a 4-class news topic classifier, using accuracy and macro F1:

| Model                               | Accuracy | F1     |
|-------------------------------------|----------|--------|
| Qwen3-0.6B-linear-3-adapters-merged | 0.5660   | 0.5612 |
| Qwen3-0.6B-svd-3-adapters-merged    | 0.5620   | 0.5597 |
| Qwen3-0.6B-ties-3-adapters-merged   | 0.5140   | 0.5039 |
| Qwen3-0.6B-dare-3-adapters-merged   | 0.5480   | 0.5415 |
| Qwen3-0.6B-slerp-3-adapters-merged  | 0.5700   | 0.5676 |
| Full_model                          | 0.5300   | 0.5412 |

SLERP is the clear winner on AG News, outperforming every other merge method on both accuracy (0.57) and F1 (0.5277) by a wide margin. Most of the merged models except TIES model outperformed the full\_model. This might be due to the fact that OpenMath and FineTome contain additional information that was not present in the AG News adapter.

The OpenMath and Finetome adapters were trained on reasoning and instruction-following data. These tasks require the model to:

- Understand nuanced semantic differences between topics
- Follow structured output formats
- Maintain coherent long-range context

When merged into the AG-news classification model, these capabilities transfer. AG-news classification is fundamentally about distinguishing semantic topic categories — World, Sports, Business, Sci/Tech. A model that has been pushed to reason carefully about content and follow instructions is better at distinguishing subtle topic boundaries than a model that only saw classification examples.

## Cosine Similarity Between Merged Deltas and AG News Deltas

To understand why SLERP performs so much better on AG News, the cosine similarity was measured between each merge method's final weight update and the original, individually trained AG News adapter's update, layer by layer:

| Layer                      | Linear | SLERP  | DARE   | TIES   | SVD    |
|----------------------------|--------|--------|--------|--------|--------|
| layers.0.mlp.down_proj     | 0.3557 | 0.5332 | 0.1628 | 0.2705 | 0.3473 |
| layers.0.mlp.up_proj       | 0.3814 | 0.5172 | 0.1769 | 0.2942 | 0.3785 |
| layers.0.self_attn.k_proj  | 0.4695 | 0.5340 | 0.2171 | 0.3751 | 0.4674 |
| layers.0.self_attn.q_proj  | 0.4274 | 0.5252 | 0.1960 | 0.3396 | 0.4235 |
| layers.0.self_attn.v_proj  | 0.3693 | 0.5272 | 0.1742 | 0.2883 | 0.3664 |
| layers.1.mlp.up_proj       | 0.4323 | 0.5332 | 0.2053 | 0.3363 | 0.4260 |
| layers.1.self_attn.v_proj  | 0.3718 | 0.5387 | 0.1826 | 0.2844 | 0.3662 |
| layers.10.self_attn.q_proj | 0.3836 | 0.5358 | 0.1807 | 0.3052 | 0.3788 |
| layers.10.self_attn.v_proj | 0.4091 | 0.5246 | 0.1985 | 0.3327 | 0.4061 |
| layers.11.mlp.down_proj    | 0.3031 | 0.5109 | 0.1394 | 0.2374 | 0.2977 |

SLERP shows the highest cosine similarity to the original AG News adapter at every single sampled layer, typically around 0.50–0.54, while DARE shows the lowest, typically around 0.13–0.22. Linear and SVD sit close together in the middle (roughly 0.30–0.47), with TIES slightly above them. This directly explains the accuracy results: the merge method that best preserves the direction of the original AG News update (SLERP) is the one that performs best on AG News, while the method that diverges from it the most (DARE) performs worst.